

MALINDI, Kenya

WMO: 637990

Lat: 3.229S

Lon: 40.102E

Elev: 25

StdP: 101.03

Time Zone: 3.00 (E03)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.0	22.8	18.1	13.1	25.6	19.0	13.8	25.4	10.9	27.0	10.2	27.1	1.8	180	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	6.0	32.2	26.0	32.1	25.9	31.8	25.8	27.1	30.6	26.7	30.2	26.6	30.1	6.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	21.6	29.4	25.9	21.3	29.2	25.3	20.5	28.8	85.6	30.8	84.2	30.6	83.2	30.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.4	8.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	27.8	28.2	28.8	28.8	26.9	26.4	25.7	25.5	26.1	26.9	27.8	28.3		
	DBStd	1.49	0.78	0.74	0.85	1.12	1.19	0.92	0.75	1.48	1.07	0.84	0.89	0.73		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6297	551	508	583	565	525	491	487	479	483	524	534	567		
	CDD18.3	3256	293	275	325	315	267	241	228	222	233	266	284	309		
	CDH23.3	33068	3146	3080	4016	3863	2547	2097	1626	1618	1844	2488	3118	3626		
	CDH26.7	9502	934	1018	1643	1555	554	300	137	161	283	595	1001	1320		
(14) Wind	WSAvg	4.6	5.2	4.9	4.0	4.7	4.8	5.4	5.4	5.2	5.0	4.2	3.1	3.8		
(15) Precipitation	PrecAvg	1042	10	2	43	173	305	166	109	65	45	93	71	32		
	PrecMax	1709	91	30	275	430	601	952	410	239	173	612	235	109		
	PrecMin	662	0	0	0	4	68	14	11	10	1	0	0	0		
	PrecStd	257	19	5	57	101	125	143	74	48	40	128	58	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	32.2	33.2	33.1	31.2	29.7	29.0	29.1	29.9	31.1	32.1	32.2		
		MCWB	25.7	25.5	26.4	26.3	25.9	24.7	23.7	23.4	24.2	24.9	25.6	26.2		
	2%	DB	31.2	31.7	32.2	32.4	30.2	29.1	28.2	28.2	29.1	30.2	31.2	31.9		
		MCWB	25.3	25.3	25.9	26.1	25.5	24.5	23.4	23.2	23.8	24.7	25.8	26.0		
	5%	DB	30.9	31.1	32.0	32.0	29.8	28.8	28.0	28.1	28.8	30.0	31.0	31.2		
		MCWB	25.2	25.1	25.9	26.1	25.3	24.3	23.3	23.2	23.6	24.6	25.9	25.9		
	10%	DB	30.2	30.8	31.6	31.5	29.1	28.2	27.2	27.8	28.1	29.2	30.2	30.9		
		MCWB	24.8	25.0	25.8	26.0	24.9	23.9	23.0	23.1	23.3	24.3	25.7	25.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	26.5	27.4	27.5	26.7	26.0	24.7	24.7	25.3	26.2	26.7	27.6		
		MCDB	30.0	30.7	31.3	30.8	29.6	28.5	27.5	27.3	28.1	29.1	30.3	30.7		
	2%	WB	26.5	26.0	26.8	26.9	26.1	25.1	24.2	24.2	24.7	25.7	26.6	27.0		
		MCDB	29.6	30.2	30.6	30.6	29.3	28.0	27.2	27.0	27.7	28.7	30.0	30.4		
	5%	WB	25.9	25.6	26.6	26.7	25.7	24.7	23.8	23.7	24.2	25.4	26.3	26.6		
		MCDB	29.2	29.7	30.3	30.3	28.7	27.7	26.9	26.7	27.3	28.3	29.6	30.0		
	10%	WB	25.3	25.2	26.3	26.4	25.3	24.4	23.5	23.5	23.9	25.0	26.0	26.2		
		MCDB	28.9	29.3	30.0	30.0	28.2	27.4	26.6	26.5	27.0	27.9	29.2	29.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	5.6	6.0	5.0	4.4	4.0	4.0	4.8	4.8	5.3	5.4	5.7		
		MCDBR	6.0	6.0	6.5	5.6	4.8	4.3	4.2	4.8	5.0	5.8	5.8	6.0		
	5% WB	MCWBR	2.3	2.0	2.3	1.9	2.2	2.4	2.3	2.4	2.1	2.1	2.2	2.4		
		MCWBR	5.8	5.7	5.9	5.1	4.6	4.1	4.1	4.7	4.9	5.2	5.5	5.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.505	0.512	0.468	0.418	0.435	0.464	0.456	0.433	0.419	0.431	0.431	0.481		
		taud	2.148	2.124	2.295	2.486	2.415	2.275	2.290	2.370	2.408	2.394	2.415	2.247		
	Ebn at Noon	Ebn at Noon	838	834	863	883	842	802	816	858	892	894	898	854		
		Edh at Noon	161	166	139	111	114	128	128	123	122	125	122	144		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.04	6.29	6.31	5.71	4.93	4.86	4.99	5.47	6.03	6.18	6.16	6.09		
	RadStd	RadStd	0.22	0.19	0.27	0.34	0.37	0.20	0.23	0.21	0.23	0.33	0.28	0.23		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		+0.36	+0.47	+0.46	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page